Healthcare planner agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number:AAI-02

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Academic Year:2025-26

Problem Statement & Objectives

1. Problem Statement

The Healthcare Planning Assistant Agent is an AI-based system designed to help users plan healthcare actions based on their symptoms and location. The main goal is to simplify decision-making by providing useful medical information, identifying nearby hospitals, and suggesting a structured treatment plan.

Objectives:

- To design an AI-based system to help users plan healthcare actions based on symptoms and location.
- To simplify decision-making by providing useful medical information and identifying nearby hospitals.
- To suggest a structured treatment plan by utilizing a multi-agent approach.

2. Scope of the Project

The system takes two inputs: the disease or symptoms being experienced and the user's location. The core scope includes:

- Implementing a multi-agent system where different agents (Planner, Research, Hospital Finder, Scheduler) collaborate to solve different parts of the problem.
- Gathering relevant medical information and identifying nearby hospitals/clinics using external tools like web search and location-based services (OpenStreetMap/Overpass).
- Combining collected data to generate a concise, structured healthcare plan.

Proposed Solution

1. Key Features

The system offers the following key features:

- Multi-Agent System (CrewAI): Uses specialized agents for research, hospital finding, and scheduling.
- Data Integration: Combines medical research from web search with nearby facility data from OpenStreetMap/Overpass.
- Structured Output: Generates a concise and structured final healthcare plan.

- Mobile-First Frontend: A production-ready React web app with a responsive UI, smooth animations, and fast feedback.

2. Overall Architecture / Workflow (CrewAI Pipeline)

The system runs a sequential CrewAI workflow that generates a concise healthcare plan from a disease/symptom topic and a city/location input.

Workflow Stages:

- **Research Task:** Gathers condition summary, symptoms, treatments, and specialist guidance via web search (DuckDuckGo Search).
- **Hospital Task:** Calls hospital_search to fetch real nearby hospitals (from OpenStreetMap/Overpass).
- **Doctor Task:** Calls doctor_search to fetch nearby doctors/clinics (from OpenStreetMap/Overpass).
- **Schedule Task:** Composes a short structured final plan.

3. Tools & Technologies Used

The project uses the following technologies:

🔹 **Backend Tech Stack:**

- Python
- CrewAI (Agent orchestration)
- LangChain Groq (LLM)
- DuckDuckGo Search (ddgs)
- OpenStreetMap APIs (Nominatim + Overpass)

Project Files (Backend):

- main.py: CLI entry point
- crew_setup.py: Crew orchestration
- tasks.py: Task definitions and expected outputs
- agents.py: Agent setup and custom tools
- requirements.txt: Python dependencies
- .env: Local secrets (not committed)

🔹 **Frontend Tech Stack:**

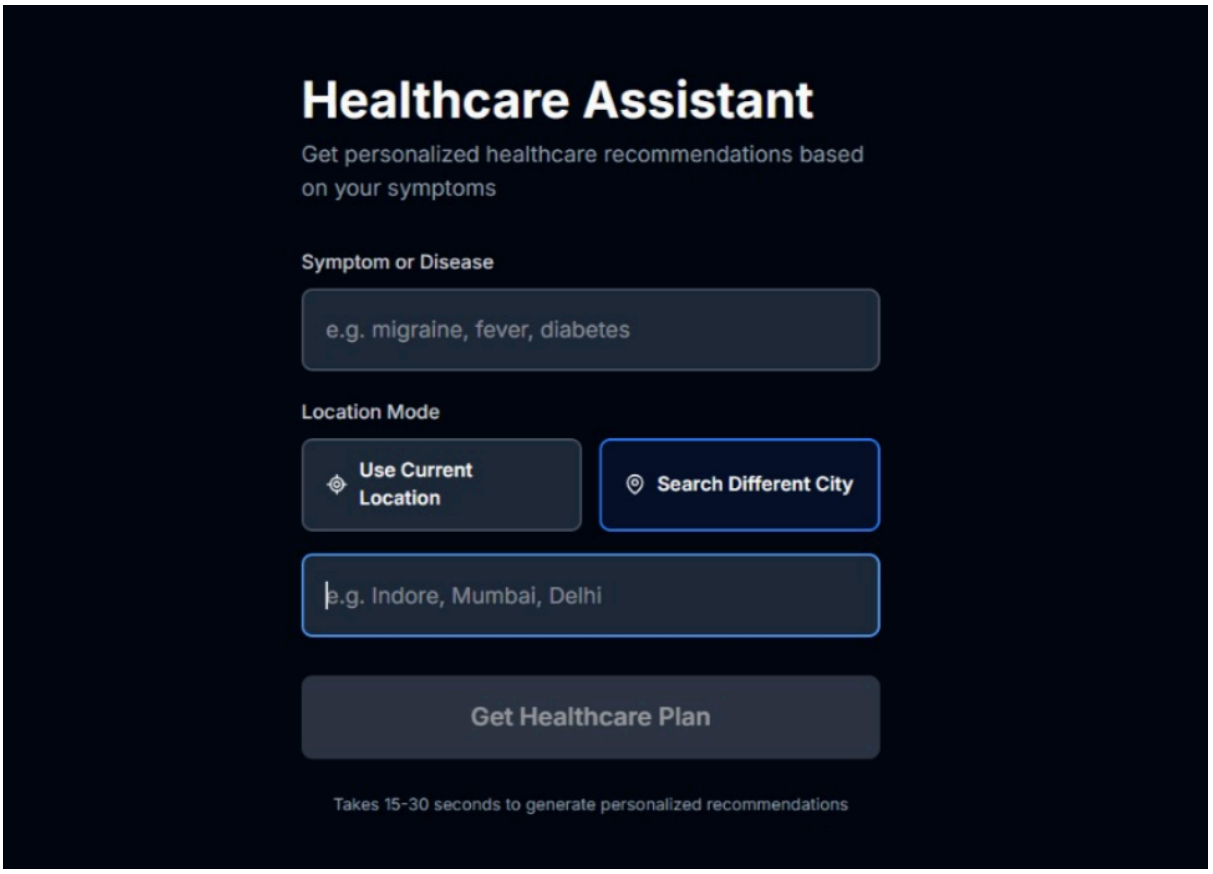
- React 18 - UI library
- Vite - Build tool

- Tailwind CSS - Styling
- React Query (TanStack) - Data fetching & caching
- Axios - HTTP client
- Lucide React - Icons

Frontend Features:

- Mobile-First Design: Responsive UI optimized for small screens first
- Fast & Smooth: Skeleton loaders, smooth animations, and optimized queries
- Real-Time Feedback: Loading states, error handling, and retry capabilities
- Caching: React Query with intelligent cache management
- Accessible: Semantic HTML, ARIA labels, keyboard navigation

4. Results & Output



Your Healthcare Plan

Final Treatment Plan for Migraine

1. Condition Summary: Migraine is a neurological disorder whose main symptom is intense headaches. There's no cure, but medical treatment and healthy lifestyle changes can help keep them under control.

2. Symptoms:

- Intense headaches
- Nausea
- Aura (affects vision)
- Sensitivity to light, sound, or smell
- Vomiting

3. Treatment Guidance:

- Manage symptoms of a migraine attack with pain-relieving medications
- Prevent migraine attacks with lifestyle changes and medications
- Combination of treatments may be necessary

4. Recommended Specialist:

- Migraine specialist
- Headache specialist

Conclusion

The Healthcare Planning Assistant Agent project successfully shows how agent-based AI systems can be used to solve real-world problems by breaking complex tasks into smaller, manageable steps and combining the results into a meaningful output. The system utilizes a multi-agent approach (CrewAI) for execution and is delivered through a high-performance, mobile-first React application.

Future Scope & Enhancements

Future Scope (Possible Extensions):

- Integrate real hospital databases for real-time hospital availability.
- Add features for direct appointment scheduling.
- Build a more user-friendly web or mobile interface.
- Address the dependency of results on the accuracy of external APIs.

Enhancements (Focus on Limitations and Frontend UX):

- Performance: Implement code splitting with lazy loading, React Query caching, and optimized re-render with memo components.
- UX Features: Implement Smart Input Validation, Skeleton Loaders, Smooth Animations, Auto-Scroll, Error Recovery, and Theme Persistence.
- Development: Auto-reload on file changes, Fast refresh with Vite, and Tailwind CSS JIT compilation.